

Conjecture 00000003482: the maximal gap $\chi - \chi_f$ is at least $n/10$
for $n = 5t$, so it is not of order $n/(\log n)^2$

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1 The conjecture

English. Definition: The chromatic scale deviation is the extremal behavior of the gap between the chromatic and fractional chromatic numbers. Conjecture: There exist graph families where the difference between the chromatic and fractional chromatic numbers exceeds $n^{1/3}$, attained by compositions of Kneser graphs with Mycielski constructions; the maximal order of deviation is n over the logarithm squared, with extremal structure pseudorandom graphs.

Chinese text. The Chinese version says the same; its last clause reads (translated) “the maximal order of the deviation is n divided by the logarithm squared, and the extremal structure is pseudorandom graphs”. The deviation is the gap (difference) between the chromatic number and the fractional chromatic number.

2 Reading

The conjecture is a conjunction. We refute its *order clause*. For a finite graph G let the deviation be $\chi(G) - \chi_f(G)$ (the text calls it the “gap” and the “difference” between the two numbers), and let

$$M(n) = \max_{G \text{ a graph on } \{0, \dots, n-1\}} (\chi(G) - \chi_f(G)).$$

“The maximal order of deviation is $n/(\log n)^2$ ” is read as a growth statement about $M(n)$ as $n \rightarrow \infty$. We refute each of the following readings:

$$(O) \ M(n) = O(n/(\log n)^2);$$

$$(\Theta) \ M(n) = \Theta(n/(\log n)^2);$$

$$(\sim) \ M(n) \sim n/(\log n)^2.$$

More generally we show that M is not $O(f)$ for any $f = o(n)$, so the base of the logarithm, or reading the expression as $n/\log(n^2)$, does not matter.

Readings not refuted. A purely lower-bound reading $M(n) = \Omega(n/(\log n)^2)$ is true (by the same family) and is not refuted. A reading in which the “deviation” is the ratio χ/χ_f rather than the difference is not addressed. The existence clause (a family with gap exceeding $n^{1/3}$) is not needed and is in fact satisfied by our family.

Conventions. $\log$ is the natural logarithm (`Real.log`); at $n \leq 1$ Lean has $\log n = 0$ and $x/0 = 0$, which is irrelevant for statements along `atTop`. O , Θ and $\sim$ are Mathlib’s `IsBigO`, `IsTheta` and `IsEquivalent` along `atTop` on $\mathbb{N}$. $\chi(G)$ is Mathlib’s `SimpleGraph.chromaticNumber` (a value in $\mathbb{N} \cup \{\infty\}$), converted to $\mathbb{N}$ by `ENat.toNat`; this is harmless because $\chi(G) \leq |V| < \infty$ for finite graphs, which we prove where used.

3 Definitions

Definition 1 (Fractional chromatic number). Let G be a finite simple graph on V . A *fractional colouring* is a function $w : 2^V \rightarrow \mathbb{R}$ with $w(S) \geq 0$ for all S , $w(S) = 0$ unless S is independent, and $\sum_{S \ni v} w(S) \geq 1$ for every vertex v . The fractional chromatic number is $\chi_f(G) = \inf\{\sum_S w(S) : w \text{ a fractional colouring}\}$.

This is the standard linear-programming definition. Wikipedia, *Fractional coloring* (https://en.wikipedia.org/wiki/Fractional_coloring): “Fractional graph coloring can be viewed as the linear programming relaxation of traditional graph coloring.”; $\chi_f(G)$ “can be obtained as a solution to a linear program”; “For each independent set I , define a nonnegative real variable x_I .” (the objective is $\sum_I x_I$ subject to $\sum_{I \ni x} x_I \geq 1$ for each vertex x). Allowing w on all subsets but forcing $w = 0$ off independent sets is the same LP. In Lean: `IsFracColoring` and `fracChromaticNumber` (an `sInf` in $\mathbb{R}$). The set of fractional colourings is nonempty (weight 1 on singletons; `fracColoring_nonempty`), so the infimum is not a junk value.

The deviation is `gap G` $= \chi(G) - \chi_f(G)$, and `maxGap n` $= \sup_G \text{gap } G$ over all $G : \text{SimpleGraph}(\text{Fin } n)$ (a finite supremum, i.e. $M(n)$).

4 The witness family

Let G_t be the join of t copies of the 5-cycle C_5 : vertex set $\{0, \dots, t-1\} \times \mathbb{Z}_5$, with $(i, a) \sim (j, b)$ iff $i \neq j$, or $i = j$ and $a - b = \pm 1$. It has $n = 5t$ vertices. In Lean, `joinC5 t` on `Fin t × Fin 5` (using Mathlib’s `cycleGraph 5`) is transported to `witness t : SimpleGraph (Fin (t * 5))` along `finProdFinEquiv`.

Lemma 2 (`three_colors`). *Every proper colouring of C_5 uses at least three colours.*

Proof. Let $c_0, \dots, c_4$ be the colours, consecutive ones distinct. If $c_0 \neq c_2$ then c_0, c_1, c_2 are distinct. Otherwise $c_2 = c_0$; if $c_3 \neq c_1$ then c_0, c_1, c_3 are distinct ($c_3 \neq c_2 = c_0$); otherwise $c_3 = c_1$ and c_0, c_1, c_4 are distinct ($c_4 \neq c_0, c_4 \neq c_3 = c_1$). $\square$

Lemma 3 (`three_mul_le_of_coloring, chromaticNumber_witness`). $\chi(G_t) \geq 3t$.

Proof. Given a proper colouring with k colours, let S_i be the set of colours used on copy i . By the previous lemma $|S_i| \geq 3$. Vertices of different copies are adjacent, so the S_i are pairwise disjoint. Hence $3t \leq \sum_i |S_i| = |\bigcup_i S_i| \leq k$. $\square$

Lemma 4 (`wt_isFracColoring, fracChromaticNumber_witness`). $\chi_f(G_t) \leq 5t/2$.

Proof. Put weight $1/2$ on each set $\{(i, a), (i, a+2)\}$ ($i < t, a \in \mathbb{Z}_5$) and 0 elsewhere. Each such set is independent, since a and $a+2$ are not adjacent in C_5 . A vertex (i, a) lies in the two distinct sets coming from a and from $a+3$ (as $a+3+2 = a$), so it receives weight at least 1. There are at most $5t$ such sets, so the total weight is at most $5t/2$. $\square$

Theorem 5 (`gap_witness, maxGap_ge`). $\chi(G_t) - \chi_f(G_t) \geq t/2$, hence $M(5t) \geq t/2 = n/10$.

5 Asymptotic conclusion

Theorem 6 (`not_isBigO_of_isLittleO`). *If $f : \mathbb{N} \rightarrow \mathbb{R}$ satisfies $f = o(n)$, then M is not $O(f)$.*

Proof. If $M = O(f)$ then $M = o(n)$. Composing with $t \mapsto 5t$ (which tends to infinity), eventually $|M(5t)| \leq \frac{1}{20} \cdot 5t = t/4$. For $t \geq 1$ this contradicts $M(5t) \geq t/2$. $\square$

Lemma 7 (`div_log_sq_isLittle0`). $n/(\log n)^2 = o(n)$.

Proof. $n/(\log n)^2 = n \cdot (\log n)^{-2}$ and $(\log n)^{-2} \rightarrow 0$ because $(\log n)^2 \rightarrow \infty$. $\square$

Corollary 8 (Main theorem).

- *maxGap_not_isBig0*: $\neg(M = O(n/(\log n)^2))$;
- *maxGap_not_isTheta*: $\neg(M = \Theta(n/(\log n)^2))$;
- *maxGap_not_isEquivalent*: $\neg(M \sim n/(\log n)^2)$.

Hence the order clause of Conjecture 00000003482 is false under readings (O) , (Θ) and $(\sim)$, and so the conjunction is false.

6 Lean formalization

File `lean/Conjecture3482/Basic.lean` (namespace `C3482`, only import `Mathlib`, 262 lines). The main statement is (Unicode symbols spelled out in ASCII here)

```
theorem maxGap_not_isBig0 :
  NOT maxGap =0[atTop] (fun n : Nat => (n : Real) / Real.log n ^ 2)
```

where `maxGap n := iSup (fun G : SimpleGraph (Fin n) => gap G)` and `gap G := (G.chromaticNumber.toNat : Real) - fracChromaticNumber G`. The file builds with `lake build`. Axiom audit (`#print axioms`) for `maxGap_not_isBig0`, `maxGap_not_isTheta`, `maxGap_not_isEquivalent` and `not_isBig0_of_isLittle0`: `[proptext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`; `decide` is used only on statements about `Fin 5`.